

PIZZA



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Welcome To Pizza

INTRODUCTION

The food service industry generates massive volumes of transactional data every day. Converting this raw data into meaningful business insights is essential for improving sales performance, customer satisfaction, and operational efficiency.

This project focuses on analyzing pizza sales data using SQL to uncover key trends, customer purchasing behavior, and revenue-driving factors. Through structured queries and data analysis techniques, the project transforms raw sales records into actionable insights that support better business decision-making.

Using SQL queries, the project demonstrates how data can be extracted, filtered, aggregated, and visualized to answer real-world business questions. The insights generated from this analysis can help restaurant management optimize inventory planning, pricing strategies, staffing decisions, and marketing campaigns.

Retrieve the total number of orders placed.

```
SELECT  
    COUNT(order_id) AS total_orders  
FROM  
    orders;
```

	total_orders
▶	21350



Calculate the total revenue generated from pizza sales.

```
SELECT
  SUM(orders_details.quantity * pizzas.price) AS total_sales
FROM
  orders_details
  JOIN
  pizzas ON pizzas.pizza_id = orders_details.pizza_id
```

	total_sales
▶	317711.60000000038



Identify the highest-priced pizza.

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

	name	price
►	The Greek Pizza	35.95



calculate top 5 pizzas by size

```
SELECT
  pizzas.size,
  COUNT(orders_details.order_details_id) AS order_count
FROM
  pizzas
  JOIN
  orders_details ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

size	order_count
L	7020
M	6097
S	5791
XL	101
XXL	8



Select top 5 pizzas along with their quantity

```
SELECT
  pizza_types.name, SUM(orders_details.quantity) AS quantity
FROM
  pizza_types
  JOIN
  pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
  JOIN
  orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

name	quantity
The Barbecue Chicken Pizza	2083
The California Chicken Pizza	1531
The Big Meat Pizza	1501
The Classic Deluxe Pizza	1331
The Hawaiian Pizza	1022



Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
    pizza_types.category,
    SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

category	quantity
Classic	6186
Chicken	5576
Veggie	3866
Supreme	3700



Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(order_time) AS hours, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

hours	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920



Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT  
    category, COUNT(name)  
FROM  
    pizza_types  
GROUP BY category;
```

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9



Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
  AVG(quantity)
FROM
  (SELECT
    orders.order_date, SUM(orders_details.quantity) AS quantity
  FROM
    orders
  JOIN orders_details ON orders.order_id = orders_details.order_id
  GROUP BY orders.order_date) AS order_quantity;
```

	AVG(quantity)
▶	60.5893



Determine the top 3 most ordered pizza types based on revenue.

```
SELECT
    pizza_types.name,
    SUM(orders_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

name	revenue
The Barbecue Chicken Pizza	36862.25
The California Chicken Pizza	26740.25
The Classic Deluxe Pizza	20763.5



Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
  pizza_types.name,
  (SUM(orders_details.quantity * pizzas.price) / (SELECT
    SUM(orders_details.quantity * pizzas.price) AS total_sales
  FROM
    orders_details
    JOIN
      pizzas ON pizzas.pizza_id = orders_details.pizza_id)) * 100 AS revenue
FROM
  pizza_types
  JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
  JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC;
```

The Barbecue Chicken Pizza	11.602424966541843
The California Chicken Pizza	8.416516740339251
The Classic Deluxe Pizza	6.535329525267492
The Big Meat Pizza	5.669292528192167
The Four Cheese Pizza	4.906273488283079
The Five Cheese Pizza	4.332230865980289



Analyze the cumulative revenue generated over time.

```
select order_date,  
sum(revenue) over(order by order_date) as cum_revenue  
from  
(select orders.order_date,  
sum(orders_details.quantity * pizzas.price ) as revenue  
from orders_details join pizzas  
on orders_details.pizza_id = pizzas.pizza_id  
join orders  
on orders.order_id = orders_details.order_id  
group by orders.order_date) as sales;
```

2015-01-01	1136.35000000000001
2015-01-02	2245.85000000000004
2015-01-03	3374.85000000000004
2015-01-04	4260.15000000000001
2015-01-05	5159.1
2015-01-06	6232.3



Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select name, revenue from
(select category , name , revenue ,
rank() over(partition by category order by revenue desc ) as rn
from
(select pizza_types.category , pizza_types.name,
sum(orders_details.quantity*pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details
on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category , pizza_types.name) as a ) as b
where rn <=3 ;
```

name	revenue
The Barbecue Chicken Pizza	36862.25
The California Chicken Pizza	26740.25
The Chicken Alfredo Pizza	10351.25
The Classic Deluxe Pizza	20763.5
The Big Meat Pizza	18012
The Hawaiian Pizza	13623.25

